Exploring Cloud Microphysics Modeling Concepts the Pythonic Way

Agnieszka Żaba, Sylwester Arabas

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scope of the paper and this talk

Introduction

1. aerosol size spectrum sampling
2. unit-aware coding and plotting
3. parcel model governing equations
4. **JIT compilation using Numba**
5. parcel ODE system definition
6. test-time dimensional analysis using Pint
7. initial condition at equilibrium
8. ODE integration and result analysis
9. Super-Droplet Method algorithm
10. SDM against analytic solution

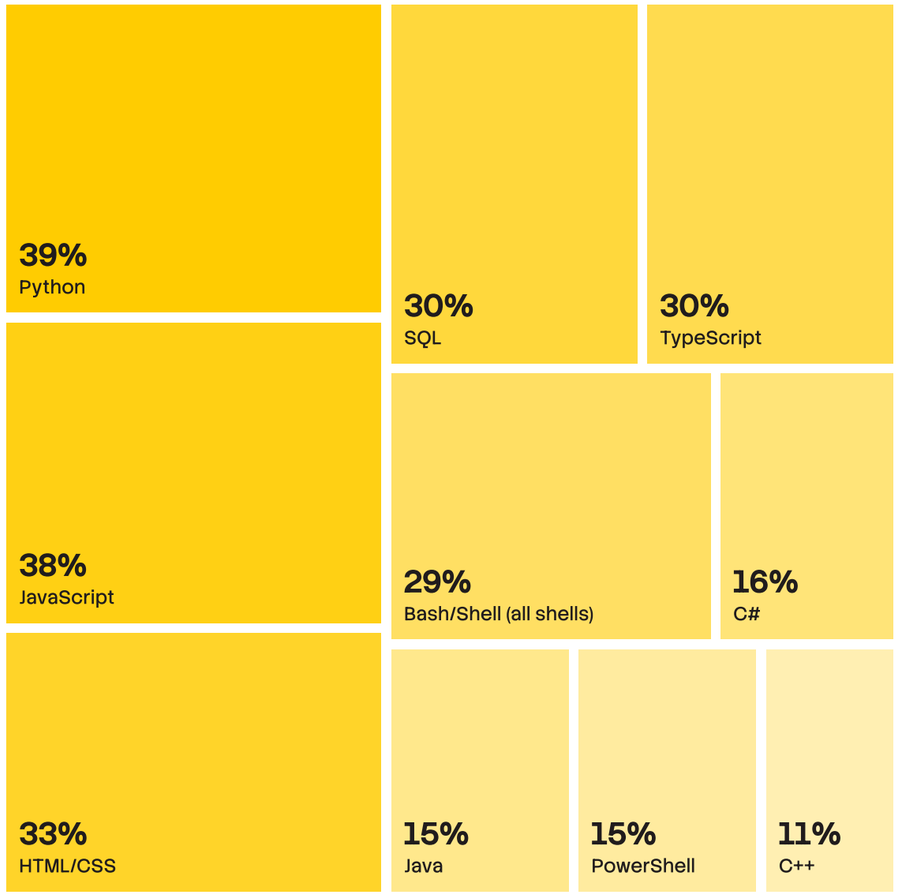
Summary

Code availability

Python and modeling?

The core of the language does not support:

- ordinary int
- multi-dimensional arrays
- JIT compilation
- multi-threading
- GPUs
- numerical algorithms



Developer Survey 2026

Technology

Most popular technologies

Top programming languages used for AI shows Python leading at 39% usage

But Python is designed as a **glue language**!

Above features are handled by packages: **NumPy**(C), **SciPy**(C/Fortran/…), **Numba/JAX**(C++/…), and many more!

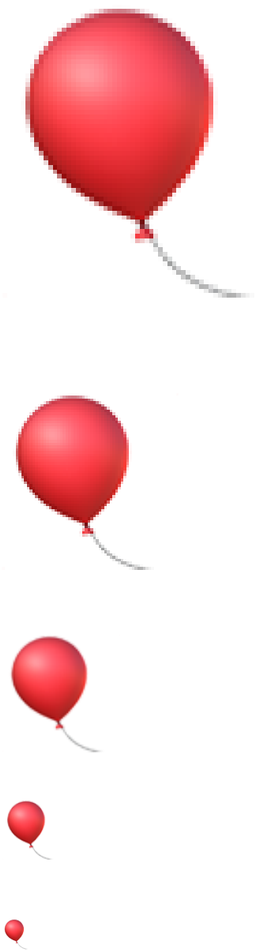
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unit-aware coding and plotting

section 2 in paper

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a useful physical example: ideal gas



equation of state linking

> ρ: density
>
> p: pressure
>
> T: temperature

$$\rho = (pM)/(TR)$$

where

> M: molar mass
>
> R: universal gas constant

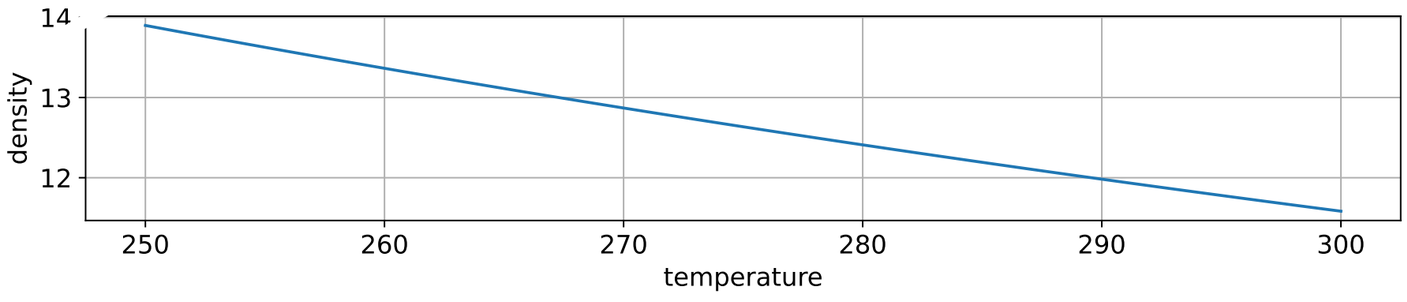
Issue: units in code (example: ideal gas density)

$$\rho = (pM)/(TR)$$

```
%config InlineBackend.figure_formats = ['svg']
import numpy, scipy, matplotlib.pyplot as plt
import mendeleev
```

```
p = 1000                                    # pressure [mbar]
T = numpy.linspace(250, 300)                # temperature [K]
R = scipy.constants.R                       # gas constant [J/K/mol]
M = (0.78 * 2 * mendeleev.N.atomic_weight + # molar mass [g/mol]
     0.22 * 2 * mendeleev.O.atomic_weight)
rho = (p * M) / (R * T)

fig, axs = plt.figure(), plt.gca()
axs.plot(T, rho)
axs.set(ylabel="density", xlabel="temperature")
```



"*The proper use of comments is to compensate for our failure to express ourself in code … Comments are always failures.*" (Robert C. Martin *Clean Code*, 2009)

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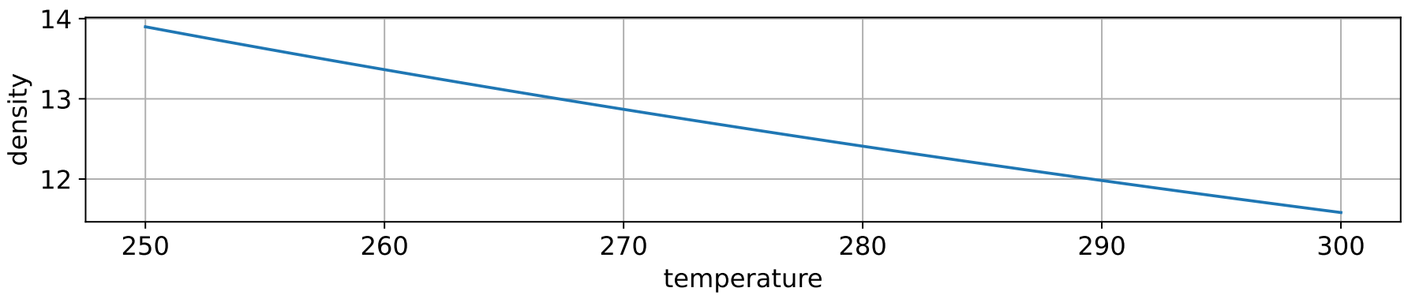
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Issue: units in comments not in code

The interpreter is not parsing the comments! We always use units on paper, use it in code as well.

- What is the unit of the plotted data?



- Reading the code, one needs to go to the definition each time.

```
1  print(T[0])
```

```
250.0
```

T is time or temperature?

- No way of enforcing changes in usage upon change of definition.
- After changing the value — did you change the comment? (We all know the answer…)

physical units in code with `Pint`

Example Gallery

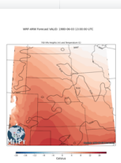
For more MetPy examples, please visit the MetPy Cookbook on Project Pythia.

General Examples

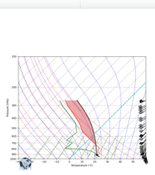
Examples of using a variety of MetPy's functionality together.



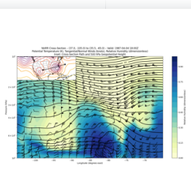
XArray Projection Handling



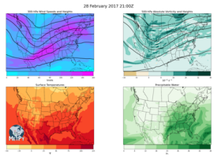
Sigma to Pressure Interpolation



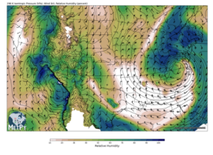
Advanced Sounding



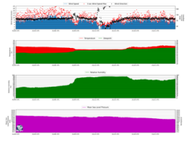
Cross Section Analysis



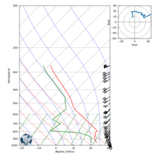
Four Panel Map



Isentropic Analysis



Meteogram



Advanced Sounding Plot with Complex

What is Pint?

Contents

- Design principles
- One last thing
- License

Pint is a Python package to define, operate and manipulate **physical quantities**: the product of a numerical value and a unit of measurement. It allows arithmetic operations between them and conversions from and to different units.

It is distributed with a comprehensive list of physical units, prefixes and constants. Due to its modular design, you can extend (or even rewrite!) the complete list without changing the source code. It supports a lot of numpy mathematical operations **without monkey patching or wrapping numpy**.

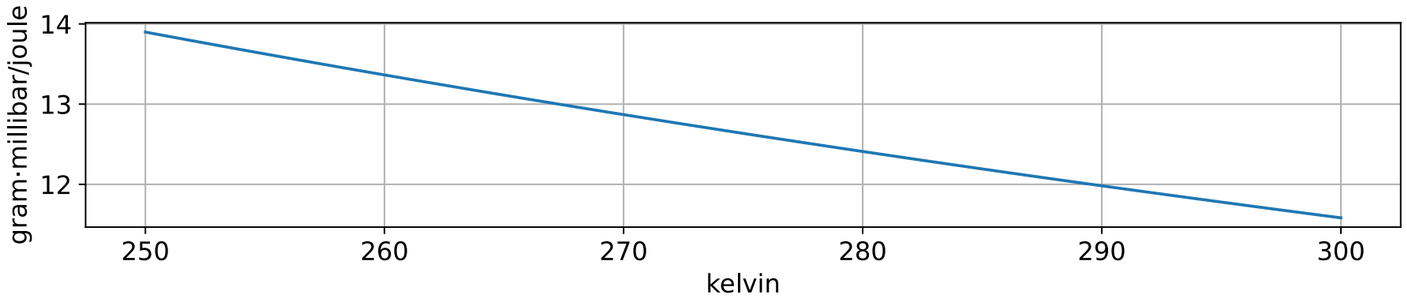
It has a complete test coverage. It runs in Python 3.12+ with minimal dependencies. It is licensed under a BSD 3-clause style license.

It is extremely easy and natural to use:

physical units with Pint (example: ideal gas density)

$$\rho = (pM)/(TR)$$

```
 1 import pint
 2 SI = pint.UnitRegistry()
 3 SI.setup_matplotlib()
 4
 5 pressure = 1000 * SI.mbar
 6 temperatures = numpy.linspace(250, 300) * SI.K
 7 gas_const = scipy.constants.R * SI.J / SI.K / SI.mol
 8 molar_mass = (0.78 * 2 * mendeleev.N.atomic_weight +
 9               0.22 * 2 * mendeleev.O.atomic_weight) * SI.g / SI.mol
10
11 density = (pressure * molar_mass) / (gas_const * temperatures)
12 plt.plot(temperatures, density)
```

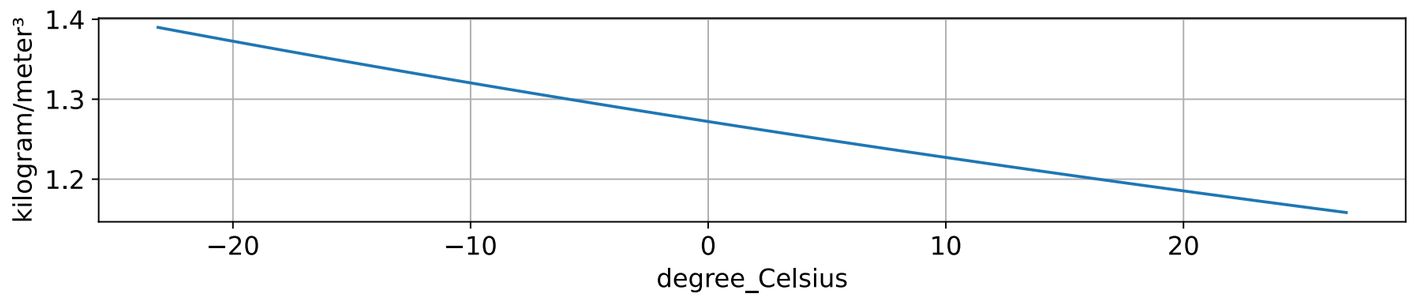


- changed names of variables
- axis labels automatic

Can we …

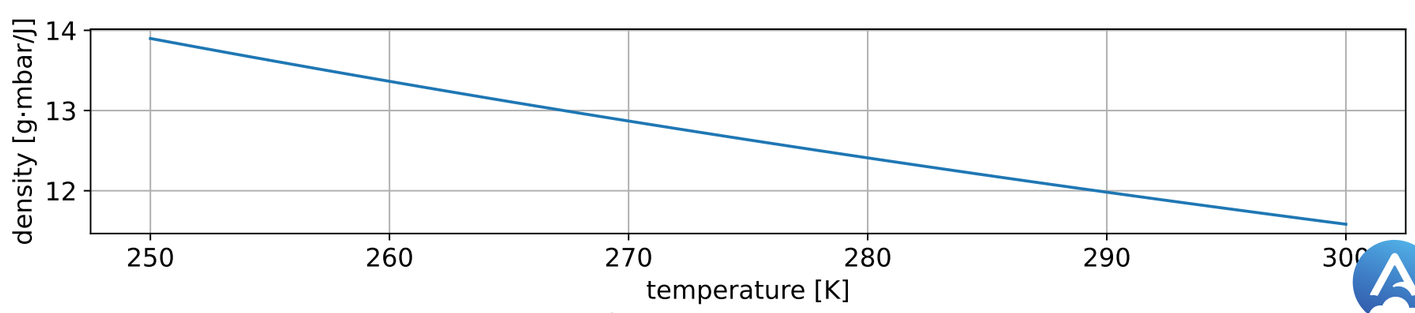
- … change units on plot?

```
1 fig, axs = plt.figure(), plt.gca()
2 axs.xaxis.set_units(SI.degC)
3 axs.yaxis.set_units(SI.kg / SI.m**3)
4 axs.plot(temperatures, density)
```



- … change axes labels and still use units?

```
1 fig, axs = plt.figure(), plt.gca()
2 axs.plot(temperatures, density)
3 axs.set_xlabel(f"temperature [{axs.xaxis.units:~P}]")
4 axs.set_ylabel(f"density [{axs.yaxis.units:~P}]");
```



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dimensional analysis using Pint

section 6 in paper

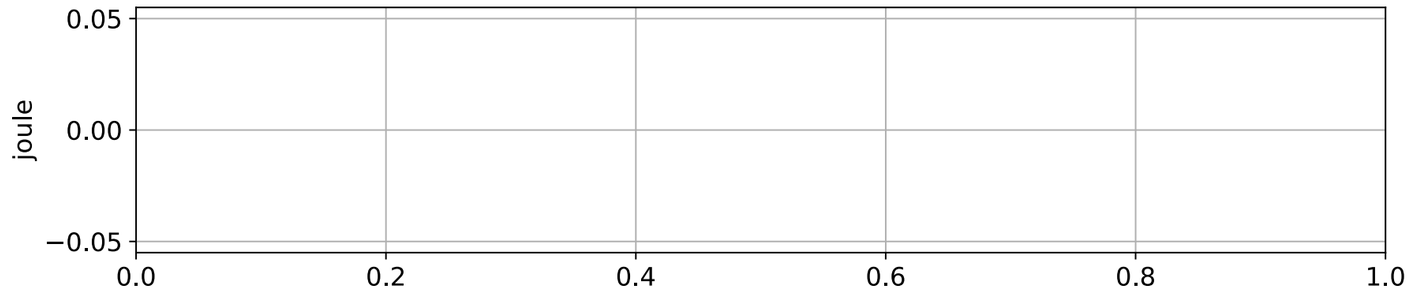
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Can we …

- … change the units to the wrong ones?

```
1 plt.gca().yaxis.set_units(SI.J)
2 plt.plot(pressure)
```

```
ConversionError: Failed to convert value(s) to axis units: <Quantity([1000], 'millibar')>
```



- … put different quantities on same axis?

```
1 try:
2     plt.plot(temperatures)
3     plt.plot(temperatures ** 2)
4 except Exception as e:
5     plt.close()
6     print(e.__cause__)
```

```
Cannot convert from 'kelvin ** 2' ([temperature] ** 2) to 'kelvin' ([temperature])
```

Can we …

- … add Kelvins to Pascals?

```
1 temperatures + pressure
```

```
DimensionalityError: Cannot convert from 'kelvin' ([temperature]) to 'millibar' ([mass] / [length] / [time] ** 2)
```

- Tools for unit testing:

```
1 ((pressure * molar_mass) / (gas_const * temperatures)).check("[density]")
```

```
True
```

```
1 assert density.check("[density]")
```

Pint: key takeaways

- ⚖️ performs dimensional analysis (unit testing!)
- ⚙️ automates plot axis labelling (and consistency checks)
- 💱 handles SI prefixes and unit conversions
- 🩺 helps in debugging (`print(x)` prints a value with unit)

Pint: key (alleged) limitations

- ⏱️ incurs performance overhead
- 🔗 no out-of-the-box interoperability with SciPy/Numba

Issue: units with SciPy and Numba

Problem with using units with SciPy/NumPy packages:

```
1 from scipy.optimize import root_scalar
2
3 f = lambda T, a, b: a / T + b
4
5 root_scalar(
6     f=f,
7     x0=1 * SI.K,
8     args=(2 * SI.K**2, -3 * SI.K)
9 )
```

```
DimensionalityError: Cannot convert from 'kelvin ** 2' ([temperature] ** 2) to 'kelvin' ([temperature])
```

Problem with using just-in-time compilation (Numba)

```
1 import numba
2
3 @numba.jit
4 def f(a, b, T):
5   return a / T + b
6
7 print(f"{f(2, -3, 270):.3g}")
```

```
-2.99
```

```
1 f(2 * SI.K**2, -3 * SI.K, 270 * SI.K)
```

```
TypingError: non-precise type pyobject
During: typing of argument at /var/folders/gl/1t3c1h9x3ws6d68b11gr2rth0000gn/T/ipykernel_58840/2583992666.py (3)
```


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Using `Pint` with `SciPy` solvers or with `Numba` JIT compiler

(`see Section 4. in the preprint`, based on BEng thesis of Daria Klimaszewska)

injectable constants-catalogue idiom

💡 key (physics) principle: dimensionality of physical quantities originates from **constants** only!

❶ ⇝ instantiate a constants catalogue at **topmost scope**, pass to all physics-related functions

❷ ⇝ use **unit-aware** constants catalogue for unit testing

❸ ⇝ use **plain-floats** constants catalogue elsewhere

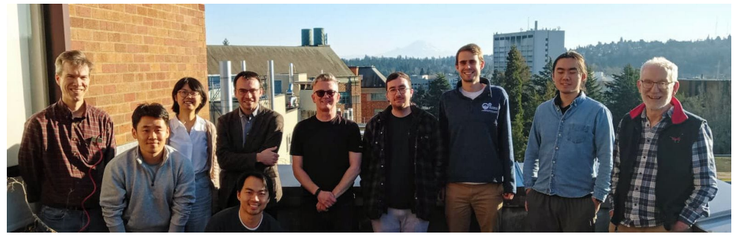
- ✅ no trade-off in **code readibility** or **performance**
- ✅ no trade-off in **functionality** (e.g., SI prefixes, unit conversions, …)
- ✅ no trade-off in **interoperability** with numerical routines or JIT compilers
- 💡 works with **any language** supporting JIT compilation (e.g., Julia) or templates (e.g., C++)
- ➕ positive side-effect: enforces **separation of concerns**

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no excuse for coding without units, let’s teach it!

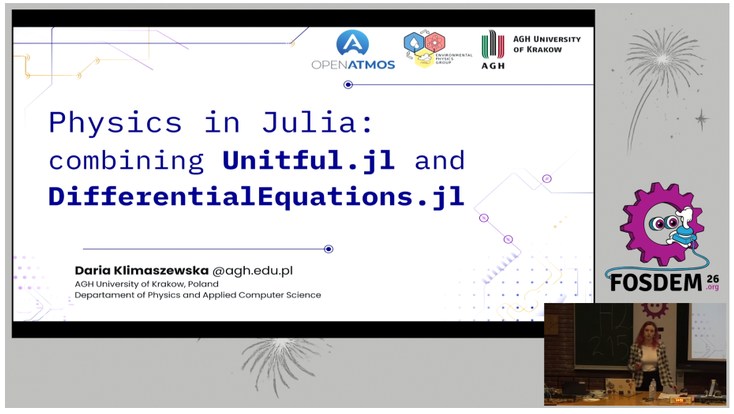
“hacking aerosol-cloud µ-physics modeling concepts in Python”

University of Washington, Jan 2026, Erasmus+ in Seattle



FOSDEM 2026 (Jan/Feb)

Université libre de Bruxelles



“Gluing SciPy, Numba, and Pint to Bridge High Performance with Maintainability”

EuroScipy 2026 (Jul, **Wydział Informatyki AGH**)



- **AGH courses**: PP, PI, MUP, MPS, RSE, MPTS, PTS…

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From executable code to a reproducible scientific paper

PySDM

Idea

- Cloud microphysics as a case study
- Executable code embedded in the paper
- Ready-to-use workshop materials
- Reproducible results and figures



Venue

- Geoscientific Model Development (GMD)
- Review and perspective paper
- GMD's reproducibility requirements

jupyter

Tools

- Jupyter for executable code
- Quarto for paper generation
- Automated tests and reproducible outputs

One executable source ⇝ Paper + Figures + Workshop

Reproducible by authors, editors, reviewers, and readers

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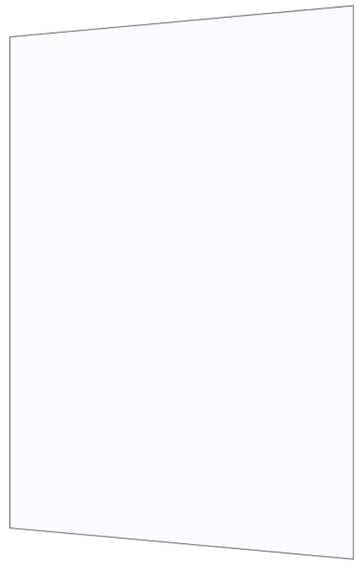
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→ Get started

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Code availability

For reproducible science use Quarto

```
quarto render paper.ipynb --execute --to copernicus-pdf
```

Exploring Cloud Microphysics Modeling Concepts the Pythonic Way

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Abstract. We share with the community a concise Pythonic codebase for hands-on exploration of cloud microphysics modeling concepts. It includes an adiabatic air-parcel model simulating diffusional droplet growth – leading to condensation, activation, evaporation, and ripening; as well as a box model resolving collisional growth of particles using the Super-Droplet Method Monte-Carlo scheme. Leveraging Pythonic abstractions and programmatic unit handling, the code combines pseudo-code-level readability, auditability, and dimensional-correctness enforcement with high performance through just-in-time (JIT) compilation, which generates native machine code on the fly, avoiding bytecode interpretation overhead for compute-heavy routines. With the goal of illustrating how concise yet complete implementations support both teaching and research software engineering, the paper includes and narrates the entire code, plotting logic included. The code runs with a single click "in the cloud" – on Google Colab or other Jupyter hubs, and thus constitutes a suitable resource for self-study or for a short course in microphysics modeling. Presented technical solutions originate from the PySDM particle-based simulation package, but are applicable in a wider scope and independently of PySDM as exemplified here. The paper includes section-wise exploratory exercises designed to guide further application of the methods and concepts discussed.

Introduction

Python in Atmospheric Sciences

In Lin (2012), there was remarkably accurate foresight predicting the widespread adoption of Python in Earth sciences. Yet, as noted in the paper, embracement of Python a dozen years ago still posed challenges, including

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- supports
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- save rendered `.tex`, `.md` files

```
1 keep-md: true
2 keep-tex: true
```

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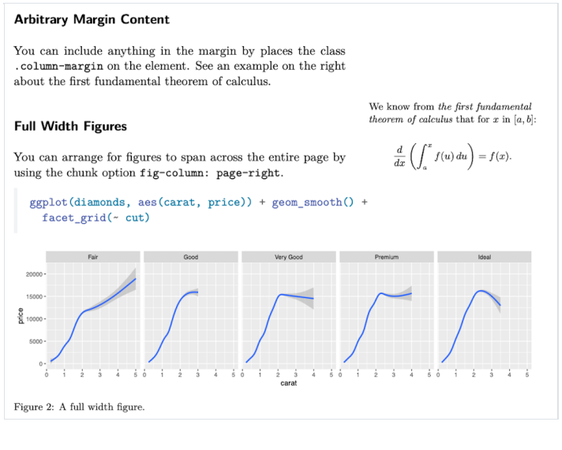
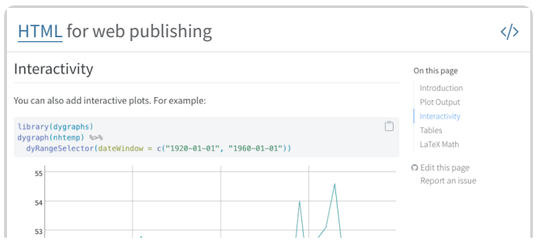


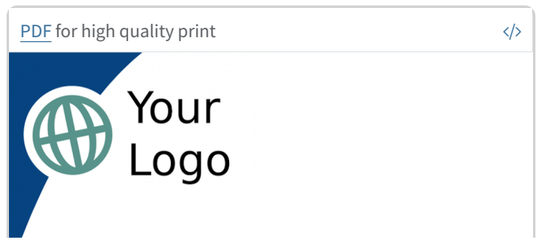
Figure 2: A full width figure.

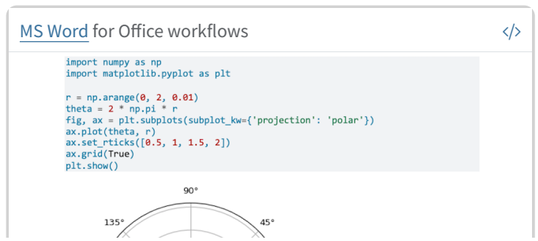
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Articles & Reports

Write a single markdown file and create output in a variety of formats.







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Quarto in Nature

nature

TECHNOLOGY FEATURE | 28 February 2022

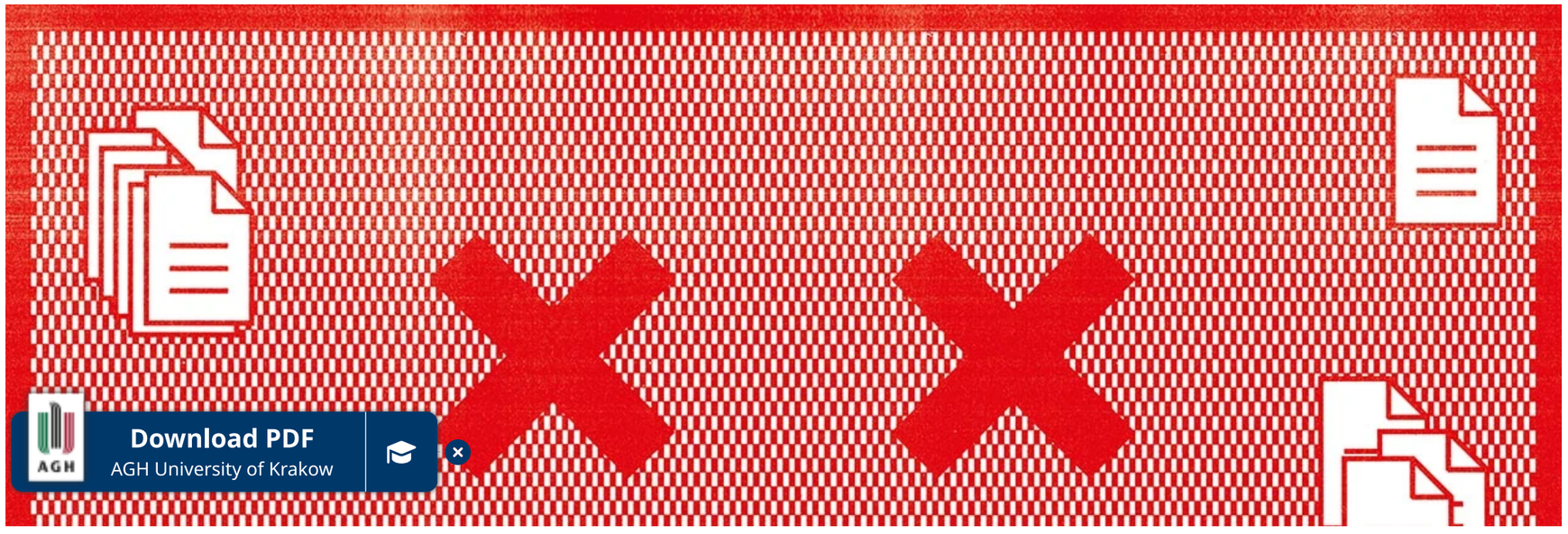
Cut the tyranny of copy-and-paste with these coding tools

‘Executable manuscripts’ insert results directly into documents, eliminating common mistakes.

By Jeffrey M. Perkel

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Quarto: Apt for BEng/MSc/PhD theses

ZFŚ Seminar, 22.10.2026

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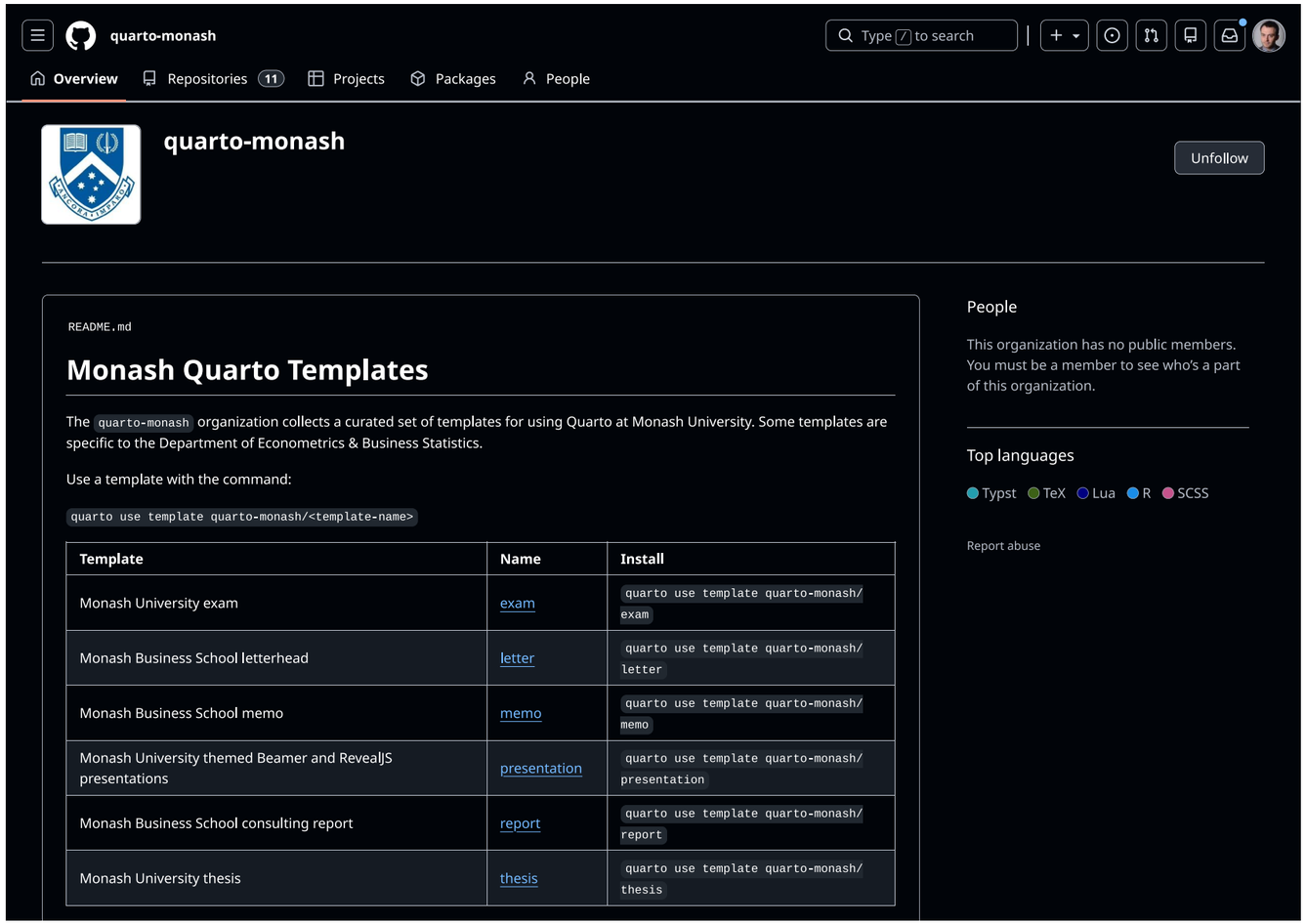
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AGH University of Krakow

University Quarto templates (theses, slides, letters)

- 🇦🇺 monash.edu (Melbourne)
- 🇳🇱 uu.nl (Utrecht)
- 🏴 ed.ac.uk (Edinburgh)
- 🇬🇧 warwick.ac.uk (Warwick)
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- 🇩🇰 au.dk (Aarhus)
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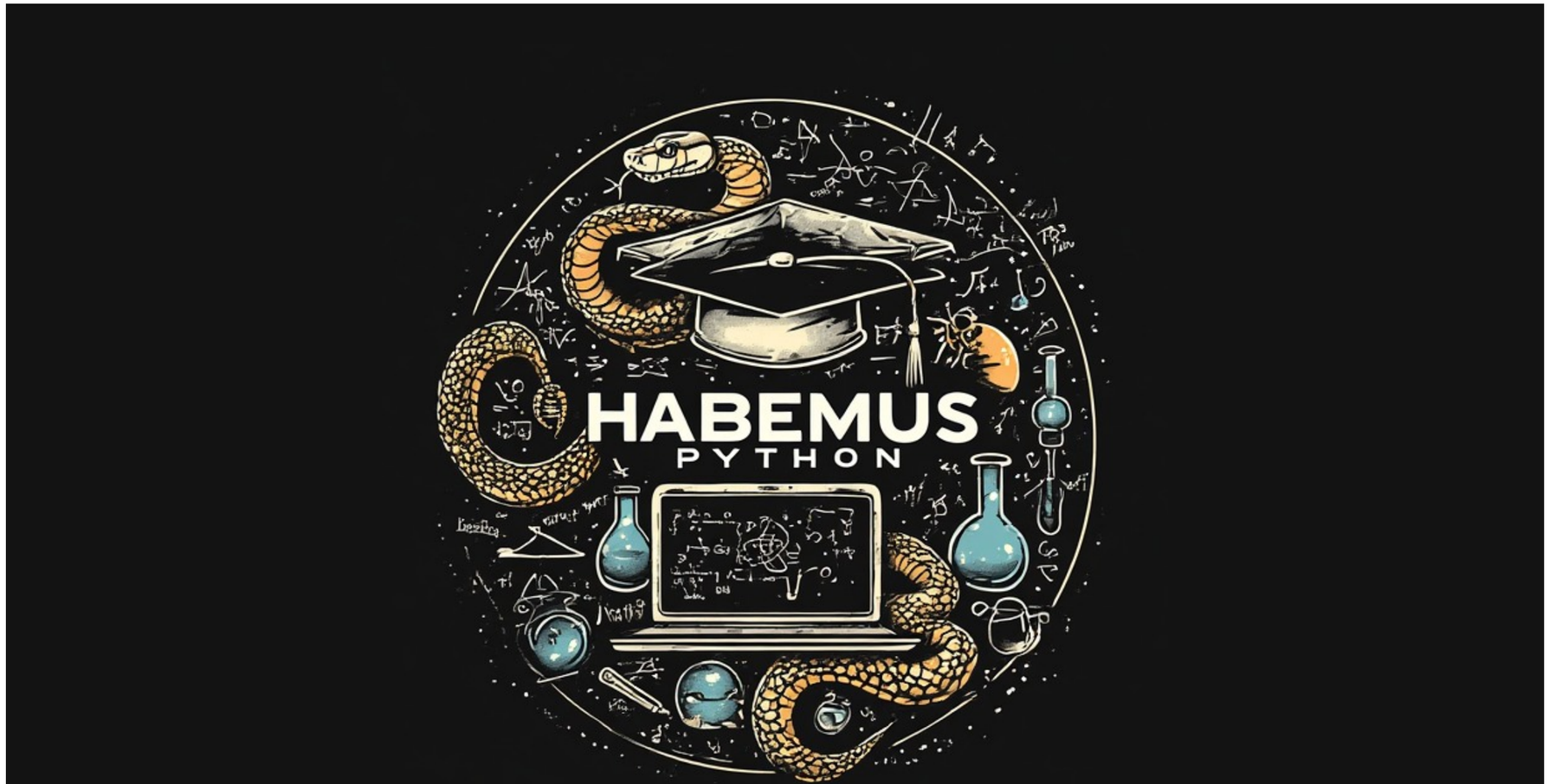


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Habemus Python https://habemus-python.github.io



Handbook of Automation for BEng & Master's projects USing PYTHON

We create resources to help students combine reproducible research, software engineering with CI automation, and high-quality, code-based typesetting. Our practical examples show how to use the "executable manuscripts" approach (see, e.g., Perkel 2022, Nature).

Write it. Run it. Share it. Engineer your documents to be reproducible, automated, and aesthetic by design!

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AKADEMIA GÓRNICZO-HUTNICZA IM. STANISŁAWA STASZICA W KRAKOWIE

WYDZIAŁ (FIKCYJNY) INŻYNIERII OPROGRAMOWANIA NAUKOWEGO

KATEDRA (FIKCYJNA) ZASTOSOWAŃ PYTHONA

Praca przykładowa

Handbook of Automation for BEng & Master's projects

Podręcznik automatyzacji prac inżynierskich i magisterskich

Autor: HABEMUS PYTHON Team
Kierunek studiów: Dowolny!
Opiekun pracy: Dr Sylwester Arabas

Kraków, 2026



ZFŚ Seminar, 22.10.2026

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Summary and outlook

- ⚖️ **Pint** (or Boost.Units in C++, UnitFull.jl in Julia, …):
  ⇝ no excuse to write code without units.
- 📝 **Quarto**:
  ⇝ highly encourage to embrace it with your mentees and mentors!
- 🌧️ **cloud/aerosol-μ-physics preprint** in public peer review in GMD:
  ⇝ DOI: `10.5194/egusphere-2026-3906` (Colab-ready Quarto/Jupyter notebook as suppl.)
- 💡 idea: **workshop/summer-school** @AGH?

Thank you for your attention!

acknowledgments: Kamil Kania & the paper co-authors

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